

How To Setup and Build a DNS Server

1. **Install the DNS Server Role:** The first step in setting up a DNS server on Windows Server is to install the DNS Server role. You can do this using the Server Manager or by using the PowerShell command-line interface.
2. **Configure Network Settings:** After installing the DNS Server role, you will need to configure the network settings for the server. This will include configuring the IP address, subnet mask, and default gateway, as well as specifying the DNS servers to use.
3. **Create a Forward Lookup Zone:** A forward lookup zone is used to resolve host names to IP addresses. You will need to create a forward lookup zone for your domain, and add resource records to the zone to define the host names and IP addresses for your network resources. You can do this using the DNS Manager snap-in in the Server Manager or by using the PowerShell command-line interface.
4. **Create a Reverse Lookup Zone:** A reverse lookup zone is used to resolve IP addresses to host names. You will need to create a reverse lookup zone for your network and add resource records to the zone to define the IP addresses and host names for your network resources. You can do this using the DNS Manager snap-in in the Server Manager or by using the PowerShell command-line interface.
5. **Configure the DNS Server Settings:** After creating the forward and reverse lookup zones, you will need to configure the DNS server settings. This will include specifying the order in which the server should use the DNS servers in your network, and configuring other settings such as caching and debugging options. You can do this using the DNS Manager snap-in in the Server Manager or by using the PowerShell command-line interface.
6. **Test the DNS Server:** Finally, you will need to test the DNS server to make sure it is working properly. You can do this by using the nslookup or ping commands from a client computer, or by using other tools such as the DNS Management snap-in in the Server Manager.